

The Macroeconomic Landscape

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Session 1

HEC Montréal | MBA

A complex and uncertain world

The Economist

"Can anything replace globalization?"

Financial Times

"Bank of Canada cuts rates as economy slows"

La Presse

"Crise du logement: le Canada n'arrive plus à suivre"

Bloomberg

"AI spending boom yet to show up in productivity data"

The Globe and Mail

"Trump tariffs threaten 75% of Canadian exports"

Wall Street Journal

"Inflation is finally cooling. Now comes the hard part."

Business implication

What do these headlines have in common? They are all **macroeconomics**. This course will give you the tools to make sense of them.

Outline

- 1. Why macroeconomics matters for business**
2. Course overview and logistics
3. Measuring the economy: GDP
4. Measuring inflation
5. Comparing living standards across countries
6. Beyond GDP
7. The macroeconomic landscape in 2026

Why Macroeconomics Matters for Business

Macroeconomics vs. microeconomics

Microeconomics provides great tools to study how individuals make decisions...
...but not how they **aggregate** to the economy as a whole
...and this aggregation, in turn, matters **a lot** for individual decision-making!

When individual rationality leads to collective madness:

- **The paradox of thrift:** everyone saves more in a recession
- **Helicopter money:** too many dollars chasing too few goods
- **The fallacy of composition:** standing up at a hockey game

Why should *you* care about macroeconomics?

Macroeconomics matters for business:

- Forecasting sales, interest rates, and wages
- Asset pricing and portfolio management
- Expanding into foreign markets and currency risks
- Anticipating regulatory and policy changes

Can you answer these questions?

1. You advise a client during a recession. Do you recommend stocks or bonds?
2. Do you expand manufacturing in Europe or Southeast Asia?
3. Do you borrow now or wait for interest rates to fall?
4. Should your firm hedge against a weaker Canadian dollar?

Course Overview

Course objectives

By the end of this course, you will be able to:

1. Understand the key **concepts and mechanisms** of macroeconomics
2. Critically interpret **macroeconomic news** (train your “BS” detector)
3. Use simple analytical frameworks to **reason about policy** and business strategy

What this course is not

Not technical. Not abstract. Not ideological.
Applied, relevant, and evidence-based.

Themes to be covered

1. **GDP and living standards** (today)
2. **Long-run economic growth:** why are some countries rich and others poor?
3. **The labor market:** inequality, technology, and AI
4. **Money, inflation, and business cycles**
5. **Monetary and fiscal policy:** how governments fight recessions
6. **Financial markets and international macro**

Logistics

Teaching material:

- Slides posted on ZoneCours before each class
- Textbook: Vincent & Yared (required readings + exercises)
- Regular reading: *The Economist*, *FT*, *WSJ*, *Bloomberg*, *La Presse Affaires*

Evaluation:

- Case study (“Global Macro Risk”): 20%
- In-class debate: 10%
- Quiz (Session 4): 10%
- Final exam: 50%
- Contribution and professionalism: 10%

Measuring the Economy: GDP

What is GDP?

Gross Domestic Product (GDP)

The total market value of all **final** goods and services produced **within a country** during a given period.

Key words:

- **Market value:** measured in dollars (or another currency)
- **Final:** excludes intermediate goods to avoid double-counting
- **Within a country:** production on Canadian soil, regardless of who owns the firm
- **During a given period:** a flow, not a stock (typically quarterly or annual)

Three ways to measure GDP

They all give the same answer

1. Expenditure

Who is buying?

$$Y = C + I + G + NX$$

- C : consumption
- I : investment
- G : gov't spending
- NX : net exports

2. Income

Who is earning?

- Wages and salaries
- Corporate profits
- Rental income
- Interest income

3. Value added

Who is producing?

- Value added at each stage
- Revenue – intermediate inputs

Key insight

Every dollar spent is a dollar earned is a dollar of value created. All three approaches yield the same GDP.

What do C, I, G, and NX include?

Consumption (C)

- Durable goods (cars, appliances)
- Non-durable goods (food, clothing)
- Services (rent, healthcare, haircuts)

Investment (I)

- Business fixed capital (machines, factories)
- Residential (new housing construction)
- Inventories (unsold goods = investment in stocks)

Government spending (G)

- Roads, defense, education, healthcare
- Does **not** include transfers (EI, pensions)
- Why? Transfers redistribute income, they don't buy goods

Net exports (NX)

- $NX = \text{Exports} - \text{Imports}$
- Can be negative (net importer)

Why do we need net exports?

C , I , and G include spending on **foreign** goods (imports).

At the same time, foreigners buy **our** goods (exports).

Since GDP measures *domestic* production, we need to correct:

$$Y = \underbrace{C + I + G}_{\text{total spending}} + \underbrace{X - M}_{\text{net exports}}$$

- Add exports (X): domestic production bought by foreigners
- Subtract imports (M): foreign production included in C , I , G

Key insight

Net exports ensure the expenditure approach captures only what is produced *within* the country.

GDP by expenditure: cross-country comparison

[Figure: GDP expenditure shares – 4 pie charts]

Canada (C 57%), US (C 68%), China (I 43%), Brazil (C 65%)

Business implication

China is investment-driven ($I = 43\%$), while the US is consumption-driven ($C = 68\%$). These differences shape trade patterns and growth strategies.

Exercise: GDP accounting

$$Y = C + I + G + NX$$

For each transaction, identify which GDP component(s) are affected and by how much.

1. A Canadian firm produces and sells a sofa for \$500. It used \$75 worth of imported parts from the US.
2. A Canadian firm produces 100 electronic components at \$80 each. The government buys 95 at \$100 each. The remaining 5 are unsold at year-end.

Hint

Think about each component separately. Remember: imports are subtracted, inventories count as investment.

Real vs. nominal GDP

Definition

Nominal GDP = output at *current* prices. **Real GDP** = output at *constant* prices.

Example: nominal GDP grew 5%, prices rose 3% $\Rightarrow$ real GDP grew $\sim 2\%$.

The **GDP deflator** captures the overall price level:

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

Key insight

Always use **real** GDP when comparing output over time. Nominal GDP confounds price and quantity changes.

Canadian real GDP since 1960

[Figure: Canadian real GDP, 1960–2025, log scale]
Shade the recessions: 1981, 1990, 2008, 2020

Canada's declining standard of living

GDP growth can mask very different **per capita** dynamics.

- Canada's real GDP per capita has **declined** relative to the U.S. since 2016
- Rapid population growth (immigration) boosted total GDP but **not per capita**
- The U.S.–Canada gap is now at its widest in decades

[Figure: Real GDP per capita]
Canada vs. U.S., 2016–2024]
Source: NBF / StatCan / BEA

Important

Always look at GDP **per capita** when assessing living standards. Total GDP can be misleading.

Can we trust GDP figures?

GDP captures, by definition, only production that can be **measured**.

- **Household production:** cooking, childcare, home repairs — not counted
- **Underground economy:** unreported income, informal work
- **Data reliability:** can we trust the numbers published by national statistical agencies?

China's GDP: how reliable?

China has quietly discontinued tens of thousands of economic statistics since Xi became leader (FT). Independent estimates suggest true growth may be much lower.

[Figure: China — declining data transparency]
Source: FT / CEIC / NBS

The importance of reliable and objective data

Data integrity is not only a concern in authoritarian regimes.

- In 2025, the U.S. administration fired the head of the Bureau of Labor Statistics
- Job growth in May–June 2025 was later **revised sharply down**
- Data revisions: –911,000 jobs over 12 months

[Figure: Data Integrity at Risk]
Monthly job changes, revised
down]

Source: NYT / BLS

Important

Reliable economic data requires **independent** statistical agencies. When independence is threatened, markets lose a critical input.

Measuring economic activity from space

[Figure: Korean peninsula at night]
South Korea lit up, North Korea dark

Satellite images of nighttime light can be used to **verify** GDP data.

Henderson, Storeygard & Weil (*AER*, 2012) show that light intensity tracks economic activity — especially useful where official data is unreliable.

Key insight

GDP data is imperfect. Always think critically about the numbers.

Measuring Inflation

What is inflation?

Inflation

The rate of change in the **overall price level** between two periods.

$$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \% \Delta P$$

where P_t is the price level at time t .

Key question: how do we measure P ?

- **GDP deflator:** prices of all goods produced domestically (broad)
- **Consumer Price Index (CPI):** prices of a basket of goods bought by consumers (most commonly reported)

The Consumer Price Index (CPI)

How it works:

- Statistical agencies (Statistics Canada, BLS) survey thousands of prices monthly
- Prices are weighted by a **fixed basket** reflecting typical consumer spending
- Categories: food, housing, transport, clothing, healthcare, recreation, etc.

Headline vs. core inflation:

- **Headline:** all items (volatile)
- **Core:** excludes food & energy (less noisy)

[Figure: Canada CPI inflation]
Headline vs. core, 2005–2026
with BoC 2% target

Headline vs. core inflation

Headline inflation includes all items — but some are very volatile (food, energy, gasoline).

Core (underlying) inflation strips out extreme price changes to reveal the **trend**.

- Bank of Canada uses CPI-trim and CPI-median
- U.S. Fed focuses on “core PCE” (excludes food & energy)

Key insight

Central banks focus on core inflation for policy decisions — headline inflation is too noisy.

[Figure: CPI and core inflation]
range, Canada, 2019–2025]
Source: Bank of Canada

Inflation: why it matters for business

Inflation affects:

- **Purchasing power:** wages may not keep up
- **Interest rates:** central banks raise rates to fight inflation
- **Contracts:** long-term deals eroded by unexpected inflation
- **Asset values:** bonds lose value when inflation rises

Business implication

In 2022–23, firms faced a key decision: pass costs to consumers, or absorb them? In 2025, tariffs brought new price pressures.

[Figure: Tariff impact on prices]

Domestic vs. imported,
2024–25]

Source: PriceStats / Cavallo
et al.

Comparing Living Standards Across Countries

GDP per capita: enormous variation

[Figure: GDP per capita (PPP) since 1950]
US, Canada, Germany, S. Korea, China, India, Nigeria]
Source: Our World in Data / Penn World Table

Comparing GDPs across countries

Why exchange rates can be misleading

Two ways to convert GDP into a common currency:

1. Market exchange rates

- Useful for comparing *economic power* (e.g., trade, geopolitics)
- Problem: very volatile, and ignores price-level differences

2. Purchasing Power Parity (PPP)

- Adjusts for differences in the cost of living
- Better measure of *living standards*
- A dollar buys much more in Lagos than in New York

China vs. United States: it depends how you measure

[GDP at market exchange rate]
US > China

[GDP at PPP]
China > US since ~2016

Which measure to use?

It depends on the question. Geopolitical influence → market rates. Living standards → PPP.

Beyond GDP

What GDP does not capture

GDP is a useful but incomplete measure of well-being. It misses:

- **Inequality:** a country's GDP can grow while most citizens get poorer
- **Non-market production:** household work, childcare, volunteering
- **Environmental costs:** pollution, resource depletion, climate damage
- **Health and well-being:** life expectancy, mental health, leisure time
- **Quality improvements:** your smartphone is infinitely better than a 2005 phone, but GDP struggles to capture this

Growth has lifted millions out of extreme poverty...

[Figure: Share of population in extreme poverty by region, 1981–2020]
Source: Our World in Data / World Bank

...and saved millions of lives

[Figure: Life expectancy at birth, selected countries, 1543–2020]
Source: Our World in Data (Riley, Clio Infra, UN)

But growth alone is not enough

Growth with rising inequality?

- U.S. real GDP per capita has roughly doubled since 1980
- But median household income has barely budged
- The gains have been concentrated at the top

For managers

Inequality shapes consumer demand, labor markets, regulation, and political risk.

[Figure: GDP per capita vs.
median income (US)]

The Macroeconomic Landscape in 2026

Where we stand today

Five themes shaping the global economy

1. **Post-pandemic inflation:** supply shocks → aggressive tightening → disinflation
2. **Interest rate regime shift:** the end of “free money” after near-zero rates
3. **AI and productivity:** will the AI revolution deliver a sustained productivity boom?
4. **Deglobalization:** reshoring, tariffs, and US-China decoupling
5. **Canada:** housing affordability, immigration reversal, and fiscal pressures

Business implication

Each theme connects directly to the analytical tools we will develop in this course.

The inflation cycle: a case study in real time

[Figure: CPI inflation, Canada and US, 2019–2026]
Annotate: supply shocks, rate hikes begin, disinflation]

Key insight

In 2021, most economists believed inflation was “transitory.” It wasn’t. Understanding *why* they were wrong is one of the goals of this course.

Interest rates: the end of free money

[Figure: Policy interest rates, major central banks, 2007–2026]
Fed, Bank of Canada, ECB, Bank of England

Business implication

Higher rates mean higher borrowing costs, lower asset valuations, and a different environment for investment and M&A.

AI and the productivity question

The big question:

Will AI deliver a new era of productivity growth, or is it mostly hype?

- **Optimists:** AI is a general-purpose technology (like electricity)
- **Skeptics:** productivity growth has been disappointing for decades despite previous tech booms
- **The data so far:** massive investment, but productivity gains not yet visible in aggregate statistics

We will revisit this in Sessions 2–3.

[Figure: U.S. labor productivity
growth]
5-year averages, 1950–2025

Deglobalization and trade tensions

The world is fragmenting:

- US-China decoupling accelerating
- U.S. effective tariff rate at highest in nearly a century ($\sim 17\%$, Smoot-Hawley levels)
- “Reshoring” and “friend-shoring” as strategy
- Supply chain resilience vs. efficiency

For business

Global supply chains built over decades are being restructured. This creates both risks and opportunities.

[Figure: Progression of Trump tariffs]

US effective tariff rate, 2025]

Source: FT / Yale Budget Lab

Canada: the tariff impact in real time

The Canadian economy **contracted** in Q2 2025, driven by a collapse in exports to the U.S.

- Employment in export-dependent sectors **fell sharply** while other sectors grew
- ~75% of Canadian exports go to the U.S.
- Bank of Canada cut rates aggressively in response

Business implication

The GDP expenditure identity in action: a drop in **NX** directly reduces **Y** . We will explore these mechanics throughout the course.

[Figure: Canada GDP
decomposition]
Q2 2025 contraction]
Source: Bank of Canada

Canada in the global picture

Key challenges for the Canadian economy:

- **Housing affordability:** prices have far outpaced incomes, especially in major cities
- **Immigration policy:** rapid population growth strained housing and services; government reversed course
- **Productivity gap:** Canadian productivity growth has consistently lagged the U.S.
- **Trade exposure:** ~75% of exports go to the U.S. — tariff risks are existential
- **Declining living standards:** real GDP per capita has fallen relative to the U.S.
- **Fiscal pressures:** pandemic deficits pushed public debt to new highs

Business implication

Many of these challenges intersect. We will study each through the lens of macro theory.

Where This Course Is Going

Connecting the themes to the course

Theme	Course sessions
Why are some countries rich?	S2: Long-run growth
AI, inequality, labor shortages	S3: Labor market
Where does inflation come from?	S4: Money & business cycles
Rate hikes, stimulus, debt	S5: Monetary & fiscal policy
Trade, exchange rates, China	S6: Financial markets & international

Key takeaways

1. **GDP** measures the total value of final goods and services produced in a country
2. The expenditure identity ($Y = C + I + G + NX$) is the backbone of macro accounting
3. Always use **real GDP per capita** over time and **PPP-adjusted GDP** across countries
4. **Inflation** (measured by CPI) tracks changes in the cost of living — focus on **core** measures for the trend
5. GDP data is imperfect: think critically about the numbers and who produces them
6. The 2026 landscape: tariff disruptions, declining Canadian living standards, AI, and the end of “free money”

Before next class

Readings:

- Vincent & Yared, Chapters 1–2

To think about:

- Pick a country. Look up its GDP per capita (PPP). What surprised you?
- Find a recent macroeconomic headline. What questions does it raise?